Team details: Team-NCC

Team leader: Suleiman Alhaji Yawale

Emails: alhajiyawalesuleiman@gmail.com

Members: Mika'il Yusuf Kunle

Mentor:

Organisation: The Nigerian Communication Commission

Submission Title: AI of Resilient Connectivity (Explainable Multi-KPI Early Warning for Mobile-Network QoS Degradation)

Executive summary

AI for Resilient Connectivity addresses a core limitation in conventional telecom Quality of Service (QoS) supervision: regulators receive large volumes of performance data, but monitoring is predominantly retrospective and often examines KPIs independently. The project therefore asks a forward-looking question: can the recent behaviour of several QoS indicators be used to identify an elevated risk of significant service degradation in the next reporting period?

The solution is an explainable, human-in-the-loop early-warning system. It ingests operator cell-level submissions, validates and preserves the raw evidence, converts tens of thousands of cell observations into trusted Operator × Day analytical records, engineers temporal and multi-KPI features, and uses supervised machine learning to estimate the probability of next-period QoS degradation. A deterministic Policy Node then converts the probability into review states — NO_DECISION, MONITOR, WATCH, REVIEW or HIGH_REVIEW — without treating the model output as an automatic regulatory finding.

The design deliberately separates three functions that are often conflated: the predictive model learns numerical QoS patterns; the Policy Node applies transparent review rules; and the Knowledge Base provides documentary explanations from approved standards, regulations and project materials. Poor-quality data can prevent a model-based view, Sandbox-trained models are isolated from real-data models, and a human analyst retains final decision authority.

Problem addressed and current solutions

Problem addressed

A real-format daily 2G submission can contain more than 80,000 cell observations. The challenge is not only scale; it is interpretation. High traffic can coexist with healthy service, while gradual and coordinated movements in Availability, CSSR, HOSR, DCR, SDCCH congestion and TCH congestion may indicate emerging stress before an operator-wide average becomes obviously poor.

QoS dimension	KPI(s)	Interpretation
Availability	Cell Availability	Higher is generally better
Accessibility	CSSR	Higher is generally better
Retainability	DCR	Lower is generally better
Mobility	HOSR	Higher is generally better
Capacity	SDCCH / TCH Congestion	Lower is generally better
Demand context	2G Erlang	Load context; neither inherently good nor bad

The regulatory value of the project therefore lies in multi-KPI context. A traffic increase accompanied by stable congestion, CSSR, DCR, HOSR and Availability may indicate a resilient network carrying additional demand. The same

traffic increase accompanied by rising congestion, falling CSSR/HOSR and rising DCR can represent a materially different condition.

Current solutions and their limitations

Current QoS supervision is typically built around dashboards, KPI thresholds, periodic submissions and analyst review. These controls remain essential because they provide transparent measurements and regulatory baselines, but they are primarily descriptive: they identify what is happening now or what has already fallen below a rule or benchmark.

- Single-KPI threshold monitoring can miss coordinated deterioration that is still individually within acceptable ranges.
- Static dashboards show current or historical status but do not, by themselves, estimate next-period risk.
- Manual correlation across traffic, availability, accessibility, retainability, mobility and congestion is time-consuming when submissions contain tens of thousands of cell records.
- Simple averages can hide weak tails of the cell distribution, which is why the prototype retains percentile features in addition to means.
- A purely threshold-driven workflow does not explicitly model temporal behaviour such as 1-day change, 3-day change, rolling context or deterioration streaks.

The proposed system does not replace these existing controls. It adds a predictive and explainable layer above them, while keeping raw evidence, conventional KPI review and human judgment available for validation.

Uniqueness of the proposed solution

From retrospective monitoring to explainable early warning

The primary innovation is the shift from retrospective, single-KPI supervision to proactive multi-KPI risk estimation. The model does not simply ask whether one threshold is crossed; it learns from the combination of current KPI levels, recent changes, rolling baselines, deterioration direction and multi-KPI stress.

Feature engineering standardises the direction of deterioration so that a positive deterioration value always means worsening, even though different KPIs move in opposite directions. For example, falling CSSR and rising DCR are both transformed into positive deterioration signals. Traffic remains a contextual variable rather than being treated as degradation by itself.

Data integrity before AI

The Collector and Preprocessor are designed as audit controls, not as hidden data-cleaning steps. The original operator workbook is archived unchanged; SHA-256 fingerprints prevent duplicate processing; missing values remain distinct from zero; impossible percentage values are excluded from clean calculations while preserved in the raw archive; and data-quality gating can return NO_DECISION.

This means the model is never allowed to create confidence from data that the pipeline has already determined to be unusable.

Distribution-aware Operator × Day representation

The first implementation intentionally predicts at Operator × Day level rather than individual-cell level. PP transforms raw Operator × Cell × Day records into one daily analytical record only after validation. In addition to means, lower-tail percentiles are retained for KPIs where higher is better and upper-tail percentiles for KPIs where higher is worse, reducing the risk that an operator-wide mean hides a deteriorating subset of cells.

Prediction, policy and knowledge remain separate

The model outputs a probability; the Policy Node deterministically maps that probability, data quality and concurrent KPI deterioration to a human-review state; and the Knowledge Base answers documentary

questions using authorised sources. A Knowledge Base answer cannot change a model probability, and a model probability cannot create a regulatory requirement.

Policy state	Rule	Operational meaning
NO_DECISION	No prediction / ineligible / quality FAIL	Do not issue a model-based view
MONITOR	$p < 0.35$	Routine monitoring
WATCH	$0.35 \leq p < 0.55$	Analyst watchlist
REVIEW	$p \geq 0.55$ unless escalated	Analyst review recommended
HIGH_REVIEW	$p \geq 0.75$ and ≥2 deteriorating KPIs	Priority analyst review

These probability thresholds are configurable Sandbox review thresholds. They are not any regulatory thresholds and must be calibrated and approved before operational use.

Mapping to standards

ITU-T Y.3172 — ML pipeline architecture

The implementation follows the logical pipeline roles defined by ITU-T Y.3172. The mapping is explicit so that each function can be inspected, tested and replaced independently.

Y.3172 element	Role in the project	Implementation
SRC	Source of data	Submission from Operators; local directories now, Box adapter later
C	Collection	Discovery, operator identity, SHA-256 deduplication, raw archive and submission registry
PP	Preprocessing	Validation, cleaning, Operator × Day aggregation, feature engineering and data-quality gating
M	Model	Chronological next-period degradation classifier using Logistic Regression / Random Forest benchmarks
P	Policy	Deterministic probability-to-review rules with human-review guardrails
D	Distribution	Routes review states to alerts and latest-status outputs
SINK	Target of ML output	Regulatory analyst dashboard and API

Policy gaps, ITU AI Readiness and NAIS mapping

ITU AI Readiness mapping

The project maps most strongly to selected dimensions and themes in ITU's AI Readiness work. The mapping below is evidence-based: only areas demonstrated by the prototype are claimed.

ITU AI Readiness area	Evidence in AI for Resilient Connectivity
Data / data service capability	Real-format operator data ingestion, schema validation, missing-value handling, provenance, data-quality gating and model-eligible coverage. **GAP: Sector-specific requirements for explainability, human review, false-alarm handling, model rejection, and operational use of probabilistic predictions still need to be defined for telecom regulation.**
Innovation ecosystem — standards	Explicit implementation of the ITU-T Y.3172 pipeline and Y.3181 Sandbox concepts; reproducible code, benchmark outputs and model registry.

ITU AI Readiness area	Evidence in AI for Resilient Connectivity
Dimension 5 — Level of Integration of AI in Workflows	AI is integrated into the QoS workflow after deterministic validation and before human review; benefits can be evaluated through early-warning lead time and analyst workload. **GAP: There is no approved telecom-regulatory benchmark defining acceptable precision, recall, false-alarm rate, missed-event rate or early-warning lead time for this use case.**
Dimension 8 — Collaboration with AI	The dashboard supports human interpretation of model risk; the analyst remains part of the decision workflow rather than being removed by automation.
Dimension 9 — Impacts of Humans in AI Integration	Domain experts define target meaning, review thresholds and final action; model alerts are designed to inform, not replace, human judgment. **GAP: The appropriate division of responsibility between AI prediction and regulatory judgment needs to be operationally defined.**
Dimension 10 — AI & Policies	The project is itself a policy-oriented Sandbox: model risk is passed through a deterministic Policy Node, and the Policy/SINK separation supports experimentation with review rules without automating enforcement.
Dimension 12 — Granular Priorities	The workflow is contextualised to Nigerian telecom QoS, local operator submission structures and regulator-specific data governance needs.
Dimension 13 — Digital Infrastructure	The use case directly addresses mobile-network resilience and depends on local compute, connectivity and data-processing infrastructure for AI deployment.

National policy alignment

Nigeria's National Artificial Intelligence Strategy (NAIS) identifies five strategic pillars. The proposed prototype contributes to these pillars to different extents, with the most direct relevance arising from telecom-sector AI adoption, responsible deployment and governance. The mapping below states only the contribution that the proposed work is designed to demonstrate during the approved prototype phase [4].

NAIS strategic pillar	Relevance to the proposed prototype	Proposed evidence during prototype phase
1. Building Foundational AI Infrastructure	The project does not seek to build national AI infrastructure, but it depends on secure compute, reliable data pipelines and local model/knowledge services as enabling infrastructure.	Document the compute, storage, ingestion and local inference requirements needed to operate the prototype securely.
2. Building and Sustaining a World-class AI Ecosystem	The standards-aligned telecom use case supports local AI capability development and creates a basis for collaboration among regulators, operators, technical experts and research communities.	Produce reusable technical documentation, code, data dictionaries and standards mappings that can support wider learning and collaboration.
3. Accelerating AI Adoption and Sector Transformation	The proposal applies AI to a defined telecommunications regulatory workflow: QoS monitoring and early-warning decision support. **GAP:** The Strategy does not prescribe a telecom-specific operational framework for predictive QoS regulation, including how regulator-held KPI data should be converted into validated AI decision-support workflows.	Demonstrate how validated operator QoS data can be transformed into explainable next-period risk indicators for analyst review.

NAIS strategic pillar	Relevance to the proposed prototype	Proposed evidence during prototype phase
4. Ensuring Responsible and Ethical AI Development and Deployment	Human oversight, data-quality gating, separation of prediction from policy action, source-grounded explanations and isolation of synthetic models are built into the design. **GAP:** Sector-specific requirements for explainability, human review, false-alarm handling, model rejection, and operational use of probabilistic predictions still need to be defined for telecom regulation.	Test NO_DECISION behaviour, explainability, human-review controls, data provenance and safeguards against autonomous regulatory action.
5. Developing a Robust AI Governance Framework	The proposed architecture includes deterministic review rules, model registries, audit trails, explicit model activation and configurable policy thresholds. **GAP:** The national framework is broad; detailed telecom-sector AI lifecycle rules for model approval, monitoring, retraining, replacement and retirement are not specified at the use-case level	Define governance ownership, model lifecycle controls, threshold approval, audit requirements and conditions for any transition from Sandbox to operational use.

Evaluation Scenarios

The evaluation is structured around a staged country/regulator pathway: establish the use case and evidence base, demonstrate the operational problem, assess technical and governance responses, and then progress to controlled deployment

Knowledge Base, Datasets, Documents and Study Evidence

Asset	What is compiled	Access	Link / status
ITU-T Y.3172	Architectural framework for ML in future networks	Public link	https://www.itu.int/rec/T-REC-Y.3172/en
ITU-T Y.3181	Architectural framework for the ML Sandbox	Public link	https://www.itu.int/rec/T-REC-Y.3181/en
Nigeria National AI Strategy (NAIS)	National AI strategy and governance context	Public link	https://ncair.nitda.gov.ng/wp-content/uploads/2025/09/National-Artificial-Intelligence-Strategy-19092025.pdf
NCC QoS regulatory / business-rule material	Applicable QoS rules, KPI definitions and methodologies used for local grounding	Public link	https://ncc.gov.ng/sites/default/files/2024-11/Legal_Business_Rules_Quality_of_Service_%28QoS%29_Business_Rules_2024.pdf

Evaluation scenarios

Scenario	Test focus	Expected system behavior
Stable / high traffic	Traffic rises while QoS remains stable	MONITOR; no false degradation escalation.
Gradual multi-KPI stress	CSSR/HOSR fall while DCR/congestion rise	Risk increases; WATCH/REVIEW/HIGH_REVIEW as evidence strengthens.
Data-quality failure	Missing/invalid/duplicate records	WARNING/FAIL; preserve raw evidence; NO_DECISION where data are unusable.
Model drift / replacement	Performance changes over time	MLFO compares/retrains models in Sandbox before activation.

Evaluation and Deployment Pathway

Step 1	Use case/pilot/feasibility/data collection: deploy the prototype in a controlled environment; ingest synthetic or approved anonymized historical QoS data; verify the end-to-end SRC→C→PP→M→P→D→SINK workflow.
Step 2	Problem emerges: identify a resilience challenge such as coordinated degradation, repeated congestion, unreliable data or model drift that threshold monitoring does not address early enough.
Step 3	Regulator/Ministry deliberation: assess models and learning approach, agentic functions, Sandbox testing, automation, privacy, responsible AI, reliability, energy efficiency, model openness, human-AI collaboration, trust/adoption, available open-source or piloted solutions, and relevant standards/policy gaps.
Step 4	Controlled deployment: move only validated components into operational use; maintain data-quality gating, model registry/versioning, explicit human review, audit logs and rollback. No model prediction becomes an autonomous regulatory action.